Arista Access Node Hardware/Environmental Fault Resolution SOP for WFD-ACC-59 at Whitefield POP

Node: WFD-ACC-59 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1824

Description

The failure of Node ID WFD-ACC-59's Access role at the Whitefield POP, attributed to a Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss), is a critical event that necessitates swift and informed resolution. This Standard Operating Procedure outlines the diagnostic and resolution steps for this specific failure domain on the Arista vendor equipment, which is characterized by its unique CLI dialect and control/data plane architecture. The underlying protocol mechanics involve the Arista EOS's (Extensible Operating System) ability to monitor and manage the ASIC memory and line card operations, as well as its provision for redundant power supply configurations. When a Hardware/Environmental Fault occurs, the Arista EOS triggers the ALARM_1824 alarm, signaling a Minor Severity event that requires immediate attention. The goal of this SOP is to identify the root cause of the fault, execute the necessary diagnostic and resolution commands, and verify the fix to ensure the Access role is restored to a stable and operational state.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and the Node ID WFD-ACC-59 is reachable via the Arista EOS CLI.

- Check the redundant link status to ensure that there are no connectivity issues.

- Dispatch remote hands to the Whitefield POP, if necessary, for physical environmental checks and repairs.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal memory parity errors, Expected Output: Memory parity errors: 0x12345678, 0x90123456, Description: Executes the Arista EOS 'show system internal memory parity errors' command to display any ASIC memory parity error occurrences, which can help identify the root cause of the fault.
```

```
Command: show hardware linecard, Expected Output: Linecard 0/0:
online, Linecard 0/1: offline, Description: Runs the Arista EOS 'show
hardware linecard' command to display the status of the linecards,
which can help determine if a linecard crash is the root cause of the
fault.

Command: show power status, Expected Output: Power supply 0: online,
Power supply 1: online, Description: Executes the Arista EOS 'show
power status' command to display the status of the power supplies,
which can help determine if a power supply redundancy loss is the root
cause of the fault.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or linecard, Command: replace hardware asic, Description: Replaces the faulty ASIC or linecard with a new one, ensuring that the new component is properly installed and configured according to the Arista EOS CLI guidelines.

Step 2: Step: Power cycle the node, Command: power-cycle, Description: Performs a power cycle of the node to reset the Arista EOS and reboot the system, which can help resolve power supply redundancy loss issues.

Step 3: Step: Verify power supply redundancy, Command: show power status, Description: Verifies that both power supplies are online and redundant, ensuring that the power supply redundancy loss issue has been resolved.

Post-Check Verification

```
Command: show system internal memory parity errors, Expected Output:
Memory parity errors: 0x, Description: Executes the Arista EOS 'show
system internal memory parity errors' command to verify that there are
no ASIC memory parity error occurrences.

Command: show hardware linecard, Expected Output: Linecard 0/0:
online, Linecard 0/1: online, Description: Runs the Arista EOS 'show
hardware linecard' command to verify that all linecards are online and
functional.

Command: show power status, Expected Output: Power supply 0: online,
Power supply 1: online, Description: Executes the Arista EOS 'show
power status' command to verify that both power supplies are online
and redundant.
```

Roll-back Procedures

- Step: Restore original configuration, Command: show running-config, Description: Restores the original configuration of the node to the previous state, ensuring that any changes made during the resolution process are reverted.

- Step: Reload the node, Command: reload, Description: Reloads the node to ensure that the Arista EOS and system software are restored to their original state.